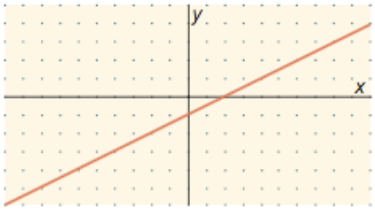
# Vectors and Parametric Equation of Lines in $R^2$

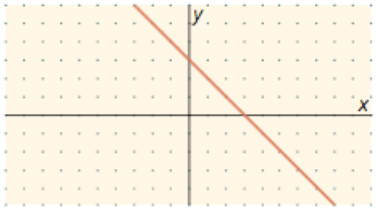
1. What is a direction vector? What is a parameter? What role do these quantities play in the equation of a line?

2. State a direction vector for each of the following lines.
   a. a line parallel to $x = 9 - 3t$, $y = -4 + t$
   b. the line through $(6, 4)$ and $(-2, -6)$
   c. the line $y = 3x + 6$
   d. a line parallel to $\vec{r} = (1, 7) + t(4, 3)$
   e. a horizontal line
   f. a vertical line

3. State the coordinates of two points on each of the following lines.
   a. $x = 3 - 8t$, $y = 4t$
   b. $\vec{r} = (4, 0) + t(0, 5)$

4. State a parametric equation and a vector equation for each of the following lines.

   a.
   
   

   b.
   
   

5. Graph the following lines.
   a. $x = -1 - 5t$
      $y = 6 + 2t$
   b. $\vec{r} = (-3, -4) + t(4, 3)$

6. For each of the following, find the parametric equations of the line that passes through the point $P$ with direction vector $\vec{d}$. In each case, find two points on the line different from $P$.
   a. $P(1, 1)$, $\vec{d} = (4, -4)$
   b. $P(5, 0)$, $\vec{d} = (1, 3)$

7. State parametric equations
   a. for the $x$-axis
   b. for a line parallel to but not coincident with the $x$-axis

8. For each of the following lines, find the vector equation that passes through the point $P$ with direction vector $\vec{d}$.
   a. $P(-2, 7)$, $\vec{d} = (3, -4)$
   b. $P\left(2, \frac{3}{4}\right)$, $\vec{d} = \left(\frac{2}{3}, 6\right)$
   c. $P(1, -1)$, $\vec{d} = (-\sqrt{3}, 3)$
   d. $P(0, 0)$, $\vec{d} = (-2, 3)$

Thinking/Inquiry/ Problem Solving

**10.** For each of the following, determine which pairs of lines are parallel and which are perpendicular.

a. $x = 1 - 3t, y = 7 + 4t$ and $x = 2 - 4s, y = -3s$

b. $\vec{r} = (1, 7) + t(-3, 4)$ and $\vec{r} = (2, 0) + s(3, -4)$

c. $\vec{r} = (1, 7) + t(-3, 4)$ and $\vec{r} = (2, 0) + s(4, -3)$

11. Find a vector equation of the line that passes through the point (4, 5) and is perpendicular to the line $\vec{r} = (1, 8) + t(3, 7)$.

12. Find the points where each of the following lines intersects the $x$- and $y$-axes. Graph the line.

a. $x = 6, y = 1 + 7t$

b. $\vec{r} = (-5, 10) + t(1, 5)$

c. $\vec{r} = (2, 3) + t(3, -1)$

Application

**13.** Show that both lines $\vec{r} = (3, 9) + t(2, 5)$ and $\vec{r} = (-5, 6) + u(3, -1)$ contain the point (1, 4). Find the acute angle of intersection of these lines to the nearest degree.

17. a. Show that $P(5, 8)$ and $Q(17, -22)$ are points on the line that passes through $A(7, 3)$ with direction vector $(2, -5)$ .

b. Describe the line segment from $P(5, 8)$ to $Q(17, -22)$ using parametric equations with suitable restrictions on the parameter.

## Answers

**2. a.** $(-3, 1)$ **b.** $(4, 5)$ **c.** $(1, 3)$ **d.** $(4, 3)$ **e.** $(1, 0)$ **f.** $(0, 1)$
**3. a.** $(3, 0), (11, -4)$ **b.** $(4, 0), (4, 10)$
**4. a.** $x = 2 + 2t, y = t$; $\vec{r} = (2, 0) + t(2, 1)$
**b.** $x = 3 - t, y = t$; $\vec{r} = (3, 0) + t(-1, 1)$
**6. a.** $x = 1 + t, y = 1 - t$; $(2, 0), (-2, 4)$ **b.** $x = 5 + t, y = 3t$; $(6, 3), (3, -6)$ **7. a.** $x = t, y = 0$ **b.** $x = t, y = 5$
**8. a.** $\vec{r} = (-2, 7) + t(3, -4)$ **b.** $\vec{r} = \left(2, \frac{3}{4}\right) + t(1, 9)$
**c.** $\vec{r} = (1, -1) + t(-\sqrt{3}, 3)$ **d.** $\vec{r} = (0, 0) + t(-2, 3)$
**9. a.** $\vec{d} = (3, -1)$ **b.** $\vec{d} = (4, 3); (5, 4)$ **c.** $\vec{d} = (1, -10); (-1, 18)$
**10. a.** perpendicular **b.** parallel **11.** $\vec{r} = (4, 5) + t(7, -3)$
**12. a.** $(6, 0)$ **b.** $(-7, 0), (0, 35)$ **c.** $(11, 0), \left(0, \frac{11}{3}\right)$ **13.** 87°
**14. a. (i)** 127° **(ii)** 11° **15. a.** $x = 24 + 85t, y = 96 - 65t$
**b.** 1h 12 min **c.** (126, 18) **16. b. (i)** $\frac{x - 5}{-8} = \frac{y + 3}{5}$
**(ii)** $\frac{x}{4} = \frac{y + 4}{1}$ **c.** $\frac{x - 7}{6} = \frac{y + 2}{1}$ **17. b.** $x = 7 + 2t$, $y = 3 - 5t, -1 \le t \le 5$ **18. b.** $0 < t < 1$ **d.** $t > \frac{1}{2}$